universität innsbruck

Department of Computer Science

# Data Preparation and Data Quality

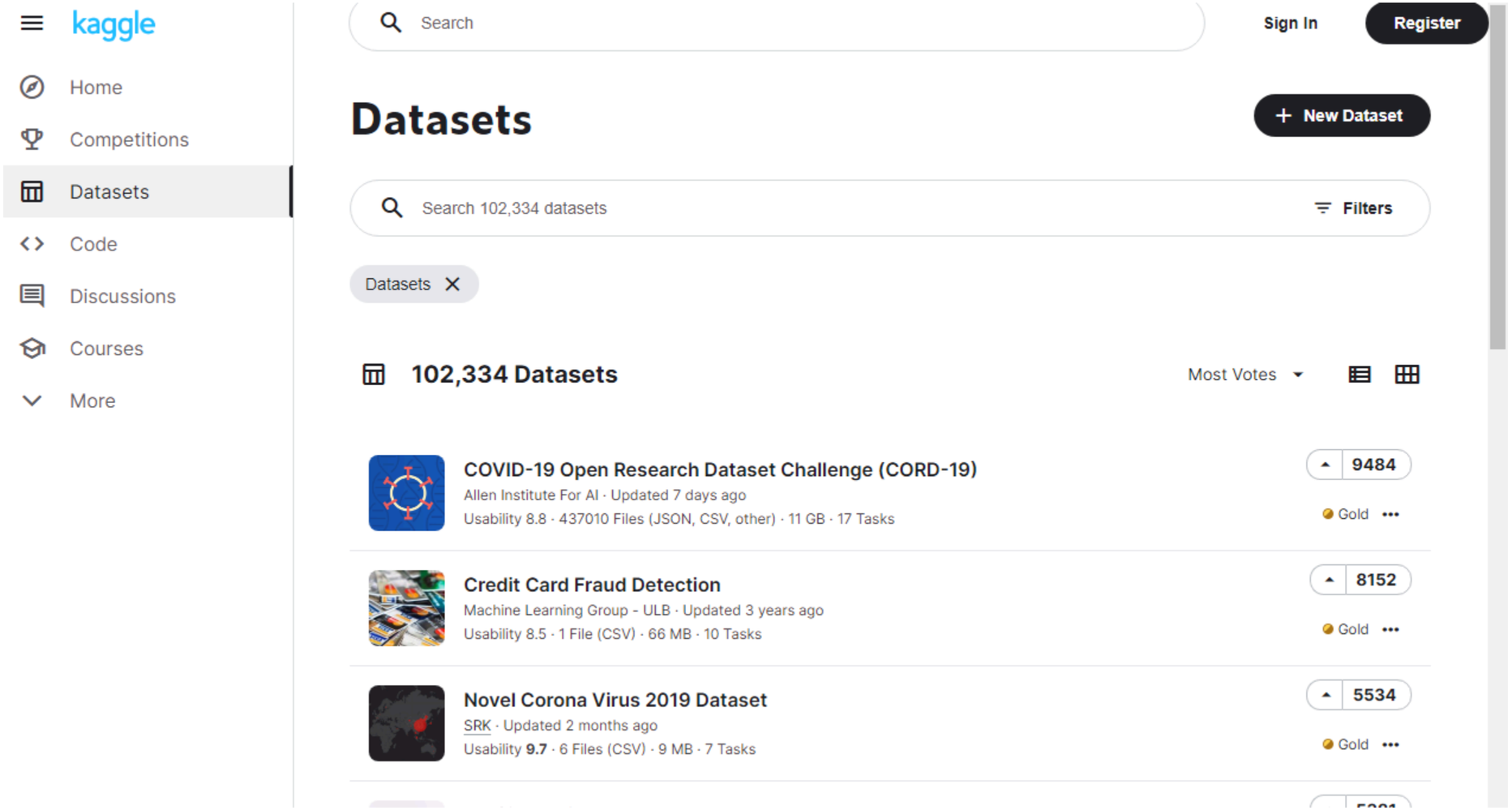
## Data Engineering and Analytics

Eva Zangerle

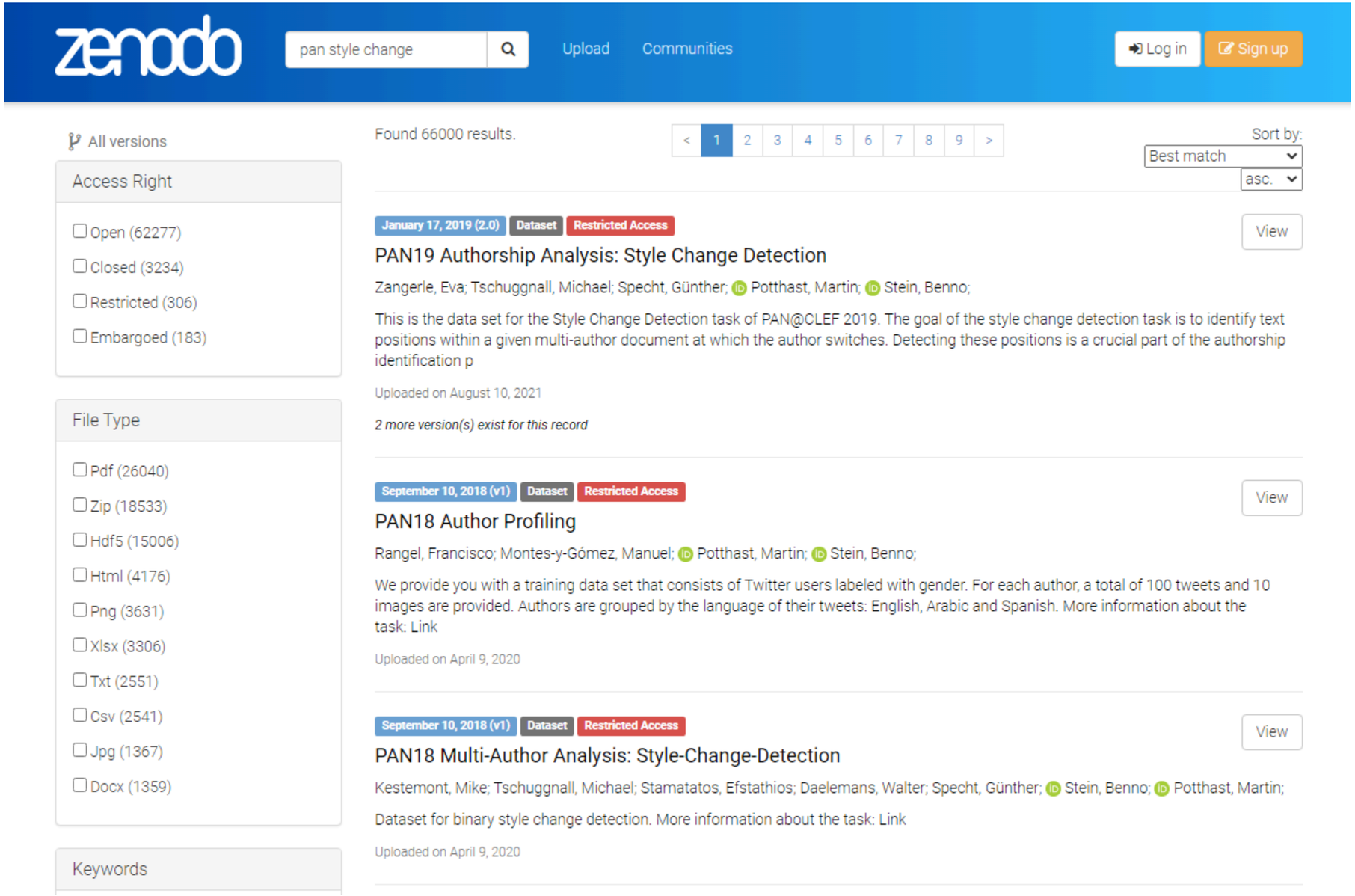
# Data Sources

- Existing datasets
- APIs
- Web scraping
- Data you are provided with in your project by business stakeholders or similar.

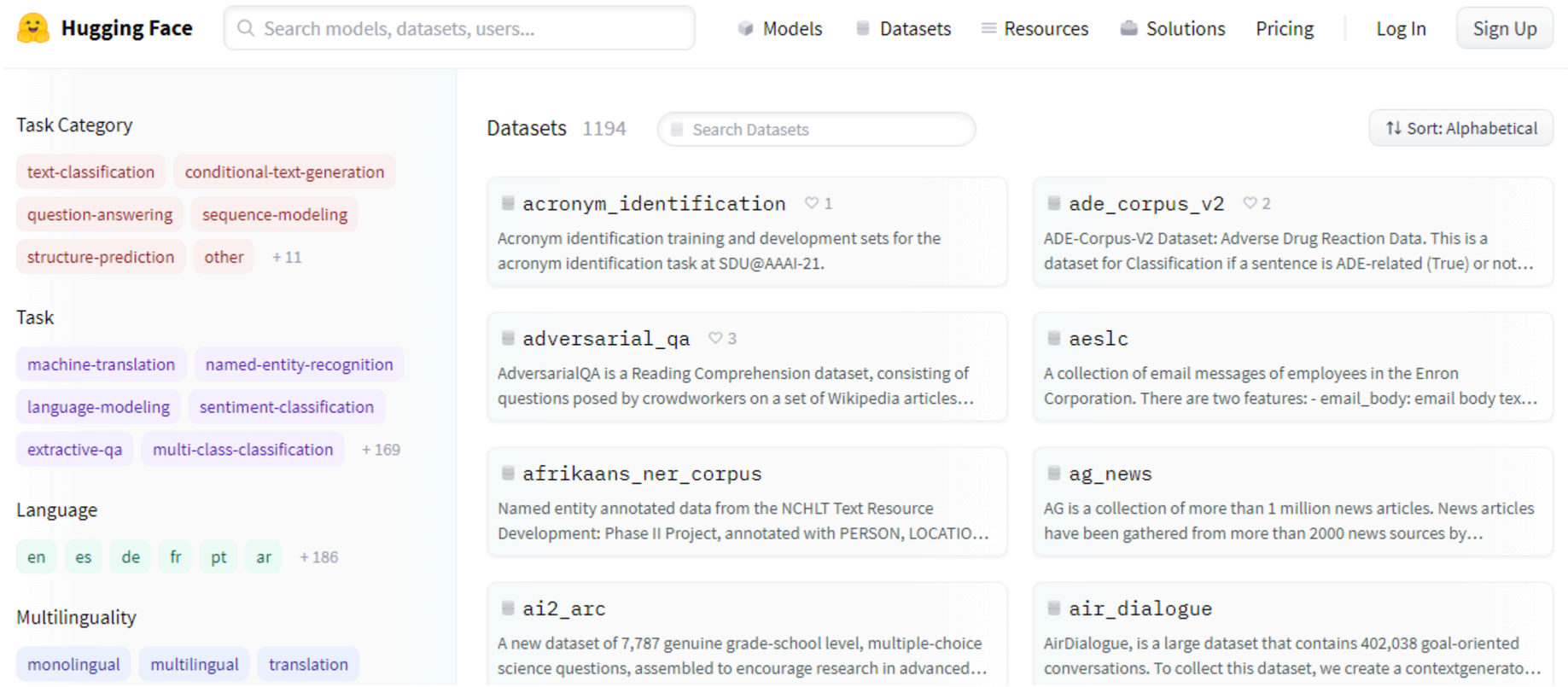
# Existing Datasets



# Existing Datasets



# Existing Datasets



# Data Ingestion

# Data Formats

- Data may come in various formats:
  - CSV
  - JSON
  - SQL access
  - NoSQL access
  - Pandas dataframes
  - Pickle files
  - etc.

Source: 03_dataset_analysis_cleaning.ipynb [CSV / JSON]

# Data Formats

- Each format may require specific handling and preprocessing steps.
- Each format may have its own problems and limitations.
- For instance, CSV parsing, escaping problems
- Often: encoding problems → individual solutions necessary

Source: 03_dataset_analysis_cleaning.ipynb [CSV / JSON]

# Data Preparation

- Goal: prepare dataset for the analysis.
- Again, it is important to know the goal of the analysis before to adapt the preparation steps accordingly.
- Main preparation steps:
  - Find anomalies in data, fix them.
  - Cope with missing data.
  - Investigate class imbalance.
  - Apply normalization and scaling.

# Anomaly Detection

# Anomalies

> Anomalies are individual data values that can be diagnosed as probably wrong
> [David Mertz, Cleaning Data for Effective Data Science]

- Anomaly detection focuses on identifying individual data points that deviate significantly from the expected pattern.
- Not to be confused with outliers, which may or may not be anomalies.
- Not to be confused with errors, which are actual mistakes in the data.
- Not to be confused with missing data, which represents the absence of a value.
- Not to be confused with data quality, which refers to the overall condition of the dataset.

# Data Anomaly - Causes

- Collection
- Measurement (instrument)
- Processing (manipulation)
- Encoding
- Transmission
- Transcription
- Data entry errors (human errors)
- Intentional errors – made deliberately to test methods
- Sampling errors
- Natural (not an error, it’s in the data)

# Dealing with Anomalies

- We distinguish between different strategies for dealing with anomalies:
  - Discard observation
    - Mark values as missing (this lecture)
  - Impute values
    - Replace with more likely values (lecture “Feature Engineering”)
- Remark: mostly, we do not actually generate the data, hence we need to deal with whatever is there.

# Anomalies: Overview

- Missing data
- Miscoded data
- Fixed bounds
- Outliers
- Multivariate outliers

# Missing Data: Overview

- SQL
- Hierarchical formats
- Sentinel values & semantics of missing data

# Sentinel Values

- Semantics of missing values (never actually collected) vs. unrepresentable/undefined/non-computable values
- Some data formats explicitly support missing data, some provide special value (sentinel value) to state missing value
- Many libraries use IEEE-754-floating point value NaN as sentinel value.
- Missing string values: mostly strings (`""`, `"N/A"`, ...)

# Missing Data: SQL

- Explicit `NULL` value.
- Nullable column: configurable for each column.
- Distinction between `NULL` and sentinel values (e.g., `NaN`) highly depends on library.

Source: 03_dataset_analysis_cleaning.ipynb [Anomaly Detection / Missing Data: SQL]

# Missing Data: Further Formats

- Hierarchical formats
  - For instance, JSON: do not represent missing data at all.
- Text formats
  - Missing values indicated by absence or sentinel value.
  - Fixed width formats: empty string, string of spaces, and missing value cannot be distinguished.
  - Two commas (no space in between) should always indicate that no value is present
  - Ideally: explicitly indicate missing values by sentinel (`Not Applicable`, `Not Calculable`, `No Middle Name`, etc.)
  - Example: Pandas 1.0 recognizes the following sentinels as missing: `' '`, `'#N/A'`, `'#N/A N/A'`, `'#NA'` ,`'-1.#IND'`, `'-1.#QNAN'`, `'-NaN'`, `'-nan'`, `'1.#IND'`, `'1.#QNAN'`, `'N/A'`, `'NA'`, `'NULL'`, `'NaN'`, `'n/a'`, `'nan'`, and `'null'`.

Source: 03_dataset_analysis_cleaning.ipynb [Anomaly Detection / Missing Data: JSON]

# Miscoded Data

- Miscoded data refers to data that has been entered incorrectly or inconsistently.
- Miscoding mostly applies to categorical data.
- Could also apply to ordinal data when boundaries are exceeded.
  - Verify type and range to assure validity.
- Quantitative data could also be entered wrongly (300 instead of 30, 5 instead of 4), tbd later.

Source: 03_dataset_analysis_cleaning.ipynb [Anomaly Detection / Miscoded Data/Ordinal Data]

# Miscoded Data

- For ordinal data, the following checks are recommended:
  - Minimum
  - Maximum of values
  - Data type to be integer/numeric format

Source: 03_dataset_analysis_cleaning.ipynb [Anomaly Detection / Miscoded Data/Ordinal Data]

# Miscoded Data

- For categorical data, the following checks are recommended:
  - Check unique values
  - Check how frequent values are
    - Rare values could indeed be improper value, but could also simply be rare values
    - Domain knowledge required to solve this issue (keep value or mark as missing)
    - In case of multiple miscoded values: which one is the correct one? (red, Red, _red?)
  - Check encoding
  - Transform/translate values accordingly

Source: 03_dataset_analysis_cleaning.ipynb [Anomaly Detection / Miscoded Data/Categorical Data]

# Fixed Bounds

- Fixed bounds are predefined limits within which data values are considered valid.
- Domain knowledge is required to estimate fixed bounds for values.
  - E.g., size of humans within certain bounds, hair length cannot be negative, etc.
  - Domain knowledge also applies to data collection procedure (e.g., how measurement was performed)
- Analyses recommended to identify and handle out-of-bounds values include:
  - Number of values exceeding bounds?
  - Distribution of those values?
  - Possible causes for out-of-bounds values?
  - Different types of errors?

Source: 03_dataset_analysis_cleaning.ipynb [Anomaly Detection / Miscoded Data/Fixed Bounds]

# Fixed Bounds

- Sometimes, looking at descriptive statistics for values exceeding bounds helps.
- For this, the 5-number summary of the sample exceeding bounds can be examined:
  - Minimum
  - Lower quartile
  - Median
  - Upper quartile
  - Maximum
- Possible solutions:
  - Drop rows
  - Impute (see later)
  - Clip values (e.g., fix all values exceeding upper bound to upper bound)

# Outliers

- An outlier is a data point that does not fall within normative range.
- It varies substantially from other values.
- Outliers can be detected using various methods, for instance:
  - Z-score
  - Interquartile range
  - Median absolute deviation
  - DBScan
  - etc.

# Z-Score

- Z-Score is a statistical measurement that describes a value's relationship to the mean of a group of values.
- It is measured in terms of standard deviations from the mean.
- A Z-Score of 0 indicates the value is exactly at the mean.
- Positive Z-Scores indicate values above the mean, while negative Z-Scores indicate values below the mean.

$$Z = \frac{X - \mu}{\sigma}$$

with X being the value, μ the mean, and σ the standard deviation.

Source: 03_dataset_analysis_cleaning.ipynb [Anomaly Detection / Outlier Detection / Z-Score]

# Z-Score cont.

- Z-Scores are most precise for normally distributed data (but usable for unimodal data (one peak), that is symmetric).
- To effectively use Z-Scores, it is important to first check if the data follows a normal distribution:
  - Visualize histogram
  - Boxplot
  - Quantile-Quantile (QQ)-plot
  - Kolmogorov-Smirnov test
  - Anderson-Darling test
  - Shapiro-Wilk test,
  - etc.

# Z-Score cont.

- 68-95-99.7 rule for normal distributions:
  - 68% of all observations within +/-1 standard deviation
  - 95% of all observations within +/- 2 standard deviations
  - 99.7% of all observations within +/- 3 standard deviations
- Rule can be used to estimate number of outliers and to tune cutoff.

# Z-Score

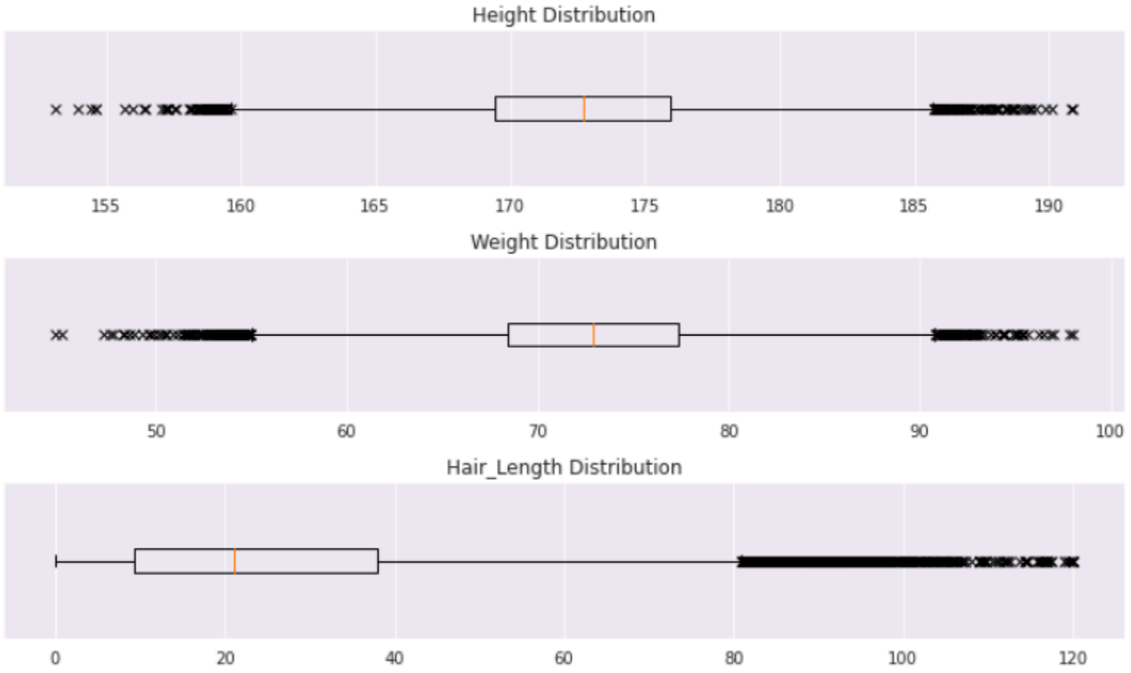
The following table illustrates the expected frequency of once-a-day observations falling outside the specified standard deviation ranges according to the 68-95-99.7 rule.

| Range | Proportion of Observations | Frequency of Once-a-Day Observations |
|---|---|---|
| > 1σ | 32% | Twice a week |
| > 2σ | 5% | Every three weeks |
| > 3σ | 0.3% | Once a year |
| > 4σ | 0.006% | Once in several decades |
| > 5σ | 0.00006% | Once in several centuries |

# Interquartile Range

- Measures dispersion/variability in data set
- Interquartile range = Q3 – Q1
  - Half of the data is in this range
  - Assumption: most of data will be within some distance beyond those Qs
  - Use a multiplier of the IQR to define this range
  - Mostly 1.5 is used → use data within $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$.
- Hardly delivers meaningful results for power law/exponential distributions

# Interquartile Range




Source: 03_dataset_analysis_cleaning.ipynb [Anomaly Detection / Outlier Detection / Interquartile Range]

# Absolute Deviation around Median

- Mean and standard deviation are sensitive to outliers
- Absolute deviation around median is more robust
- Measures dispersion/variance of data similarly to IQR
- Median Absolute Deviation from median (MAD):

$$MAD = b\, M_i \left( |x_i - M_j(x_j)| \right)$$

  where:

  - $M_i$ is median of series
  - $x_j$ original observations
  - $b = 1.4826$

- Exclude values beyond

$$M - 3\, MAD < x_i < M + 3\, MAD$$

- Values of 3 is considered highly conservative, 2.5 moderately conservative, 2 poorly conservative.

Christophe Leys, Christophe Ley, Olivier Klein, Philippe Bernard, Laurent Licata: Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median, Journal of Experimental Social Psychology, Volume 49, Issue 4, 2013, Pages 764-766.

# Exponential/Power-Law Distributions

Widely used approach: Exclude data beyond 99th/99.9th percentile.

Five-Number-Summaries of (Left) the #NP560k Data Set and (Right) the #NP90k Data Set

| | #NP560k | | | | | #NP90k | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Characteristics | Median | Q3 | Max | Mean | SD | Median | Q3 | Max | Mean | SD |
| Listening events per user | 2.0 | 4.0 | 69,197.0 | 59.83 | 1,125.48 | 2.0 | 4.0 | 463.0 | 9.16 | 31.70 |
| Listening events per track | 2.0 | 8.0 | 1,821.0 | 11.05 | 33.65 | 1.0 | 2.0 | 1,031.0 | 2.72 | 10.42 |
| Tracks per user | 2.0 | 4.0 | 69,197.0 | 59.83 | 1,125.48 | 2.0 | 4.0 | 463.0 | 9.16 | 31.70 |
| Distinct tracks per user | 2.0 | 4.0 | 3,500.0 | 10.95 | 78.73 | 2.0 | 4.0 | 319.0 | 6.26 | 19.03 |
| Hashtags per user | 2.0 | 5.0 | 86,855.0 | 74.10 | 1,446.10 | 2.0 | 5.0 | 1,025.0 | 10.84 | 39.27 |
| Distinct hashtags per user | 1.0 | 3.0 | 207.0 | 2.65 | 5.41 | 1.0 | 3.0 | 108.0 | 2.57 | 4.43 |
| Hashtags per track | 1.0 | 1.0 | 6.0 | 1.24 | 0.47 | 1.0 | 1.0 | 6.0 | 1.17 | 0.44 |
| Distinct hashtags per track | 1.0 | 1.0 | 6.0 | 1.23 | 0.46 | 1.0 | 1.0 | 6.0 | 1.16 | 0.44 |

*We show (from left to right) the median, 3rd quantile (Q3), maximum, mean, standard deviation (SD) of the individual characteristics for both data sets. The minimum and 1st quantile for both data sets are all ones for all the characteristics.*

Example taken from E. Zangerle, C. -M. Chen, M. -F. Tsai and Y. -H. Yang, "Leveraging Affective Hashtags for Ranking Music Recommendations," in IEEE Transactions on Affective Computing, vol. 12, no. 1, pp. 78-91, 1 Jan.-March 2021, doi: 10.1109/TAFFC.2018.2846596.

# Data Quality

# Data Quality

- “fit for its intended use in operations, decisions and planning”
- Data quality is context-dependent and varies according to the specific requirements of the task at hand.
- Data quality has many dimensions (see next slide), including accuracy, completeness, consistency, timeliness, and relevance.
- Consists of objective measurements and subjective perception of stakeholders.
- “Experience suggests a “one size fits all” set of metrics is not a solution. Rather, assessing data quality is an on-going effort that requires awareness of the fundamental principles underlying the development of subjective and objective data quality metrics.”

Leo L. Pipino, Yang W. Lee, and Richard Y. Wang. 2002. Data quality assessment. Commun. ACM 45, 4 (April 2002), 211–218. DOI:https://doi.org/10.1145/505248.506010

# Data Quality Dimensions

| Dimensions | Definitions |
|---|---|
| Accessibility | the extent to which data is available, or easily and quickly retrievable |
| Appropriate Amount of Data | the extent to which the volume of data is appropriate for the task at hand |
| Believability | the extent to which data is regarded as true and credible |
| Completeness | the extent to which data is not missing and is of sufficient breadth and depth for the task at hand |
| Concise Representation | the extent to which data is compactly represented |
| Consistent Representation | the extent to which data is presented in the same format |
| Ease of Manipulation | the extent to which data is easy to manipulate and apply to different tasks |
| Free-of-Error | the extent to which data is correct and reliable |
| Interpretability | the extent to which data is in appropriate languages, symbols, and units, and the definitions are clear |
| Objectivity | the extent to which data is unbiased, unprejudiced, and impartial |
| Relevancy | the extent to which data is applicable and helpful for the task at hand |
| Reputation | the extent to which data is highly regarded in terms of its source or content |
| Security | the extent to which access to data is restricted appropriately to maintain its security |
| Timeliness | the extent to which the data is sufficiently up-to-date for the task at hand |
| Understandability | the extent to which data is easily comprehended |
| Value-Added | the extent to which data is beneficial and provides advantages from its use |

# Data Quality vs. Data Cleaning

- Data cleaning focuses on detecting and correcting problems in individual observations.
- Data quality considers whether the dataset is fit for its intended purpose.
- A dataset can be free of obvious errors but still have poor quality:
  - Outdated observations
  - Unrepresentative samples
  - Missing relevant attributes
  - Insufficient coverage of the target population
- Data quality assessment should precede decisions about cleaning.
- Important: high data quality is always relative to a specific use case.

# Data Quality Assessment

- Goal: systematically assess whether a dataset satisfies defined quality requirements.
- Typical steps:
  - Define intended use and quality requirements.
  - Profile the available data.
  - Define measurable quality indicators.
  - Compare measurements against requirements.
  - Identify and prioritize quality problems.
  - Decide whether data is fit for purpose.
- Not all quality problems can be fixed by modifying the dataset.

# Data Profiling

- Data profiling: systematic examination of the structure, content, and relationships within a dataset.
- Typical analyses:
  - Number of observations and attributes
  - Data types and schema
  - Missing values per attribute
  - Unique values and duplicate observations
  - Minimum, maximum, quantiles, distributions
  - Relationships and dependencies between attributes
- Goal: obtain an initial understanding of the dataset and identify potential quality problems.
- Important: descriptive statistics alone cannot establish correctness.

# Measuring Data Quality

- Data quality dimensions can be operationalized using quantitative metrics.
- Examples:
  - Completeness: proportion of expected values present
  - Consistency: proportion of records satisfying consistency rules
  - Timeliness: age of observations or delay in data availability
  - Accuracy: proportion of values matching a trusted reference
- Example: completeness of an attribute

$$C = 1 - \frac{N_{\text{missing}}}{N_{\text{expected}}}$$

- Further quality checks:
  - Number of duplicate records
  - Proportion of values outside expected ranges
- Metrics must be interpreted with respect to the intended use.
- High scores in one dimension do not imply high overall quality.

# Limitations of Data Quality Metrics

- Quantitative metrics provide useful indicators, but cannot fully describe data quality.
- Examples:
  - Completeness: 100% of recorded values are present, but entire records may be missing.
  - Accuracy: difficult to assess without a trusted reference.
  - Consistency: values may be internally consistent but factually incorrect.
  - Timeliness: recent data is not necessarily accurate or complete.
- Quality metrics may depend on:
  - The intended use of the dataset
  - Domain knowledge
  - Availability of reference data
  - Choice of thresholds
- Important: high quality scores do not automatically imply fitness for purpose.

# Data Quality: Completeness

- Completeness is not limited to missing values within existing records.
- Distinguish:
  - Attribute completeness: are required values present?
  - Record completeness: are all expected observations present?
  - Population coverage: are relevant entities or groups represented?
- Example: all recorded transactions have complete attributes, but transactions from one store are missing.
- Consequently, a dataset can have 100% attribute completeness and still be incomplete.
- Important: missing records are often more difficult to detect than missing values.

# Data Quality: Consistency

- Consistency: absence of contradictions between related data values.
- Different forms:
  - Within-record consistency
  - Between-record consistency
  - Cross-table consistency
  - Cross-system consistency
- Examples:
  - Order total differs from sum of individual items.
  - Customer has different birth dates in two systems.
  - Foreign key references a non-existing customer.
- Recommended checks:
  - Functional dependencies
  - Referential integrity constraints
  - Cross-field validation rules
  - Comparison of redundant information

# Data Quality Rules

- Data quality requirements can be expressed as explicit, testable rules.
- Typical categories:
  - Schema rules: required columns and data types
  - Domain rules: permitted values and ranges
  - Uniqueness rules: identifiers must be unique
  - Relationship rules: foreign keys must reference existing records
  - Business rules: domain-specific constraints
- Examples:
  - `customer_id` must not be NULL.
  - `order_id` must be unique.
  - `quantity` must be greater than zero.
  - `delivery_date >= order_date`.
- Rules should be documented and automatically evaluated where possible.

# Data Quality Monitoring

- Data quality is not a static property of a dataset.
- Quality can change due to:
  - Changes in source systems
  - Modifications of collection procedures
  - Changes in the underlying population
  - Software updates and schema changes
- Quality metrics should therefore be monitored over time.
- Typical monitoring indicators:
  - Missing value rates
  - Duplicate rates
  - Validation failure rates
  - Record counts and data freshness
  - Changes in attribute distributions
- Important: changes in distributions may indicate data problems or genuine changes in the underlying population.

# Data Quality: Handling Problems

- Quality problems require different responses depending on their causes and consequences.
- Possible strategies:
  - Correct erroneous values using trusted sources.
  - Remove or impute problematic observations where justified.
  - Retain valid but unusual observations.
  - Reject or quarantine records violating critical constraints.
  - Improve data collection or processing procedures.
  - Reject datasets that are not fit for the intended purpose.
- Prioritize problems according to their impact on the intended analysis.
- Document quality checks, detected problems, and corrective actions.
- Goal: not perfect data, but data of sufficient and demonstrable quality.